

Distributions — Center, Spread, and Shape

Answer Key

Precalculus / Statistics

Quick Answers

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|-------------------|-----------|-------------------|
| 1. 65 | 5. — | 9. $\frac{7}{40}$ |
| 2. 44 | 6. 71, 71 | 10. — |
| 3. 66.67 | 7. 22, 12 | |
| 4. 59, 71.5, 12.5 | 8. -9, 40 | |
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Curriculum

Level: Precalculus / StatisticsHSS-IC.A–HSS-IC.B – *problem 10*HSS-ID.A–HSS-ID.C – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9**Difficulty 1–5 of 5 across 15 tagged checks.*

Common wrong answers

3. *If they got 74: averaged only the smallest and largest values instead of all twelve*
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The argument this sheet is making: centre and spread are not two numbers to compute and move on from — they are a *choice*, and the shape of the distribution makes it. Class A and Class B were built to have almost the same interquartile range and wildly different ranges, so that a student who reports whichever statistic came first gets a defensible answer for one class and a badly misleading one for the other. Grade problems 5, 7, 8 and 10 on whether the shape is named and used, not on whether the adjective is right.

1. Class A — the median.

With $n = 12$ the median sits between the 6th and 7th ordered values. Counting in: 52, 55, 58, 60, 62, 64, 66, 68, 70, 73, 76, 96.

$$\frac{64 + 66}{2}$$

The median is 65.

Why those two: an even count has no single middle value, so the median is defined as the midpoint of the two that straddle the middle. Note that the extreme score of 96 had no influence on this answer at all — it is simply “the largest”, and could have been 960 without moving the median. That resistance is the property problems 5 and 10 come back to.

2. Class A — the range.

Range is the largest value minus the smallest:

$$96 - 52$$

The range is $\boxed{44}$.

The value doing the work: the 96. Without it, the largest score is 76 and the range collapses to $76 - 52 = 24$ — nearly half. The range is computed from exactly two observations, both of them the most extreme in the data set, which makes it the least resistant measure of spread there is.

3. Class A — the mean.

Add all twelve scores, then divide:

$$52 + 55 + 58 + 60 + 62 + 64 + 66 + 68 + 70 + 73 + 76 + 96 = 800$$

$$\frac{800}{12} = 66.666\dots$$

Rounded to the nearest hundredth, the mean is $\boxed{66.67}$.

Compare with problem 1: the mean sits 1.67 points *above* the median. Ten of the twelve students scored below the mean. That is not an error — it is what a long right tail does, and problem 5 asks the student to say so.

Watch for: averaging only the smallest and largest value, $(52 + 96)/2 = 74$. That is the *midrange*, not the mean, and on skewed data it is even less trustworthy than the mean — it uses only the two most extreme observations.

4. Class A — quartiles and the IQR.

Split the twelve ordered values into two halves of six.

(a) Lower half: 52, 55, 58, 60, 62, 64. Its median is $\frac{58 + 60}{2}$, so $Q_1 = \boxed{59}$.

(b) Upper half: 66, 68, 70, 73, 76, 96. Its median is $\frac{70 + 73}{2}$, so $Q_3 = \boxed{71.5}$.

(c) $\text{IQR} = 71.5 - 59$, so the IQR is $\boxed{12.5}$.

What the IQR is: the width of the middle half of the data. Between Q_1 and Q_3 sit exactly six of the twelve students. Like the median, it never saw the 96: Q_3 is computed from the 9th and 10th values, so the top two observations could be anything at all.

5. Class A — centre and shape together.

(a) The mean is 66.67 and the median is 65, so

$$\boxed{66.67 > 65}$$

(b) The distribution is **skewed right**, with an **outlier** at 96.

The reasoning to require: the mean is a balance point and moves toward whichever side has values far from the centre; the median is a counting point and only cares about order. So when the mean sits above the median, the far-from-centre values are on the high side — a right tail. The dot plot confirms it directly: eleven scores are packed between 52 and 76, and one sits alone at 96, twenty points clear of its nearest neighbour.

Grading note: “mean > median so it is skewed right” is the right shape from the right evidence, and is worth full credit; the reasoning above is what you want if a student asks why the rule works. Reject “skewed right because the biggest number is on the right” — every ordered data set has its biggest number on the right.

6. Class B — a symmetric distribution.

(a) The twelve Class B scores are evenly spaced by 2, from 60 to 82. Their total is 852:

$$\frac{852}{12}$$

The mean is $\boxed{71}$.

(b) The middle two values are 70 and 72:

$$\frac{70 + 72}{2}$$

The median is $\boxed{71}$.

(c) The two centres are identical. When mean and median agree, neither tail is pulling harder than the other, which is the signature of a **symmetric** distribution — and the dot plot shows it: the points are evenly spread with no gap and no isolated value.

Why it happened here and not for Class A: Class B has no observation far from the rest. Every score is within 11 points of the centre. Class A has one at 96, thirty-one points above its own Q_1 , and the mean — which weights every observation by its actual value — gets dragged toward it while the median does not.

7. Two measures of spread, disagreeing.

(a) Range: $82 - 60$, so the Class B range is $\boxed{22}$.

(b) Q_1 is the median of 60, 62, 64, 66, 68, 70, which is 65; Q_3 is the median of 72, 74, 76, 78, 80, 82, which is 77. So $IQR = 77 - 65 = \boxed{12}$.

(c) Side by side:

	range	IQR
Class A	44	12.5
Class B	22	12

By range, Class A looks twice as spread out as Class B. By IQR, the two classes are almost identical.

The IQR is telling the truth about the bulk of the data. It measures the width of the middle half, so it is computed entirely from observations near the centre and is unaffected by how far the extremes reach. The middle half of Class A spans 12.5 points and the middle half of Class B spans 12 — the two classes really are alike in the region where nearly everyone sits.

Why the range disagrees. It is computed from two observations, and one of them is Class A's 96. Remove that single score and Class A's range drops to 24, within two points of Class B's. A statistic that one observation out of twelve can double is describing that observation, not the class.

What to report: both, honestly — “the two classes have similar middles (IQR 12.5 and 12) but Class A has one score far above the rest”. That sentence is more informative than either number alone, and it is the habit problem 10 asks for.

8. The commute histogram — reading the shape.

(a) The tallest bar is the third one, so the modal interval is $\boxed{20 \text{ to } 29}$ minutes, holding 14 workers.

(b) Add every bar:

$$2 + 8 + 14 + 9 + 4 + 2 + 1$$

The sample holds $\boxed{40}$ workers.

(c) The distribution is **skewed right**: the bars rise quickly to a peak at 20–29 minutes and then fall away slowly, with a long thin tail running out to 60–69. Because that tail is on the high side, it will pull the mean above the median — the same relationship problem 5 established for Class A, now read off a shape rather than computed from two numbers.

The check worth mentioning: the total in (b) is not decoration. Every frequency question on a histogram needs the sample size, and a student who cannot produce it has not finished reading the display — problem 9 needs it immediately.

9. The commute histogram — a tail probability.

“At least 40 minutes” means the 40–49, 50–59 and 60–69 intervals — the three bars in the right tail:

$$4 + 2 + 1 = 7 \text{ workers out of } 40$$

$$P(\text{at least 40 minutes}) = \boxed{7/40}$$

This is already in lowest terms, since 7 is prime and does not divide 40. As a decimal it is 0.175, so about one worker in six.

Reading it back into shape: seven workers out of forty spread across thirty minutes of the range, against fourteen packed into a single ten-minute interval. That imbalance — few observations covering a wide stretch on one side — is precisely what “skewed right” describes.

Watch for: including the 30–39 bar, which gives 16/40. “At least 40” excludes anything below 40, and the 30–39 interval lies entirely below it.

10. Challenge: reporting the right summary. (Open synthesis — read the reasoning.)

The rule the student should arrive at: **mean and standard deviation for roughly symmetric data; median and IQR when the data is skewed or has an outlier**, because the mean and the standard deviation both weight every observation by its distance from the centre and so are pulled by extremes, while the median and IQR are computed from position and are not.

Class A — median and IQR. The distribution is skewed right with an outlier at 96. The mean of 66.67 sits above ten of the twelve scores, so quoting it as “the typical score” misdescribes the class; the median of 65 has six scores on each side. The standard deviation would be inflated by the same single value that doubled the range, so the IQR of 12.5 is the honest width.

Class B — mean and standard deviation. The distribution is symmetric with no outlier, mean and median both 71. Here the mean loses nothing and gains something: it uses every observation rather than just the middle ones, so it is the more precise summary, and the standard deviation is likewise trustworthy when no value is dragging it.

Commute times — median and IQR. The histogram is clearly skewed right, with a tail running out to 60–69 minutes. The mean commute would be pulled above the 20–29 interval where most workers actually sit.

Full credit requires, for each of the three: (i) the shape named, (ii) the choice, and (iii) what would go wrong with the other choice, in terms of extremes pulling the mean or inflating the standard deviation. A student who writes only “Class A is skewed so use the median” has (i) and (ii) but not (iii), and (iii) is where the understanding lives.

Do not accept “always use the median because it is safer”. On symmetric data the mean is the better estimate, and a blanket rule is the absence of the judgement the problem is testing.

Open synthesis — open response.